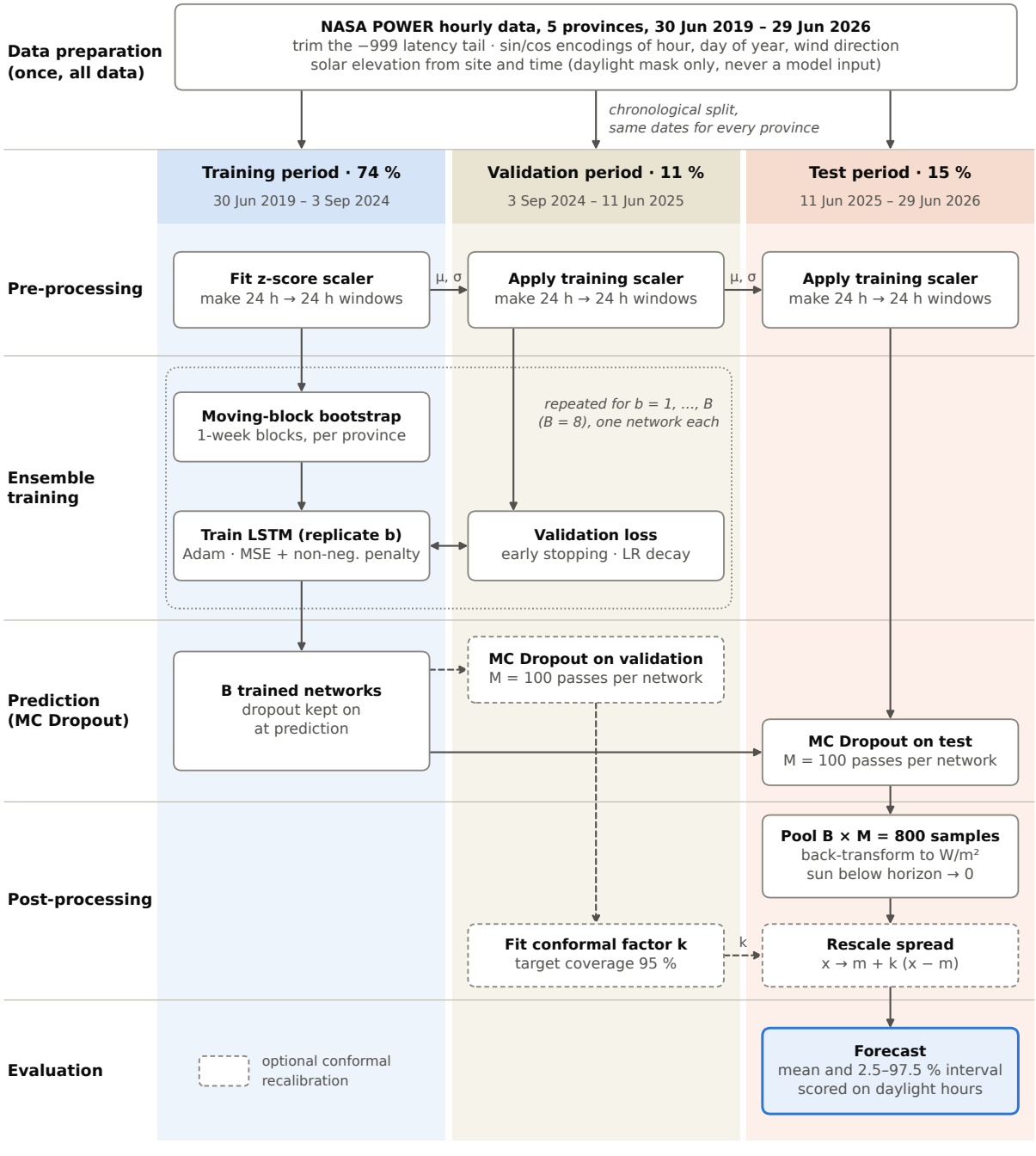

Data preparation (once, all data)
NASA POWER hourly data, 5 provinces, 30 Jun 2019 – 29 Jun 2026
trim the −999 latency tail · sin/cos encodings of hour, day of year, wind direction
solar elevation from site and time (daylight mask only, never a model input)
chronological split,
same dates for every province
Training period · 74 %
30 Jun 2019 – 3 Sep 2024
Validation period · 11 %
3 Sep 2024 – 11 Jun 2025
Test period · 15 %
11 Jun 2025 – 29 Jun 2026
Pre-processing
Fit z-score scaler
make 24 h → 24 h windows
μ, σ
Apply training scaler
make 24 h → 24 h windows
μ, σ
Apply training scaler
make 24 h → 24 h windows
Ensemble training
repeated for b = 1, …, B
(B = 8), one network each
Moving-block bootstrap
1-week blocks, per province
Train LSTM (replicate b)
Adam · MSE + non-neg. penalty
Validation loss
early stopping · LR decay
Prediction (MC Dropout)
B trained networks
dropout kept on
at prediction
MC Dropout on validation
M = 100 passes per network
MC Dropout on test
M = 100 passes per network
Post-processing
Pool B × M = 800 samples
back-transform to W/m²
sun below horizon → 0
Fit conformal factor k
target coverage 95 %
k
Rescale spread
x → m + k (x − m)
Evaluation
optional conformal
recalibration
Forecast
mean and 2.5–97.5 % interval
scored on daylight hours